

Scalable metagenomic sequence analysis using k-mer-based methods

ISMB 2026

Where to find all material

<https://github.com/bo1929/ISMB-2026-Metagenomics-Tutorial>



Schedule

- 9:00 am - 9:45 am (15 mins): Intro + background on *k*-mer-based approaches, [Siavash Mirarab](#)
- 9:45 am - 10:45 am (1 hour): **Kraken2** - [Ben Langmead](#)
- 10:45 am - 11:00 am (15 mins): Coffee Break
- 11:00 am - 12:00 pm (1 hour): **VirFinder** and *d2* - [Fengzhu Sun](#)
- 12:00 pm - 1:00 pm (1 hour): **sourmash** and **YACHT** - [David Koslicki](#)
- 1:00 pm - 2:00 pm (1 hour): Lunch Break
- 2:00 pm - 3:00 pm (1 hour): **sylph** and **skani** - [William Yu](#)
- 3:00 pm - 4:00 pm (1 hour): **krepp** - [Ali Osman Berk Şapcı](#)
- 4:00 pm - 4:15 pm (15 mins): Coffee Break
- 4:15 pm - 4:45 pm (30 mins): **DecoDiPhy** - [Shayesteh Arasti](#)
- 4:45 pm - 5:15 pm (30 mins): **kf2vec** - [Eleonora Rachtman](#)
- 5:15 pm - 5:45 pm (15 mins): Closing remarks and survey on the effectiveness of the course

Format

- Each present focuses on one or two tools. They will cover:
 - Background information about what the method does and how it works
 - A hands-on exercise
- Hands-on exercises will have different formats: some via an interactive notebook, others just as a demo

Share Your Feedback

Scan the QR code to let us know your thoughts on this tutorial.



Everyone who completes the survey will be entered into a drawing to receive a discount code for up to \$250 off an ISCB conference of their choice.

Thank you!

Quick overview of k -mers and their application in mixture analysis

Siavash Mirarab
University of California, San Diego

k -mers: Substrings of length k

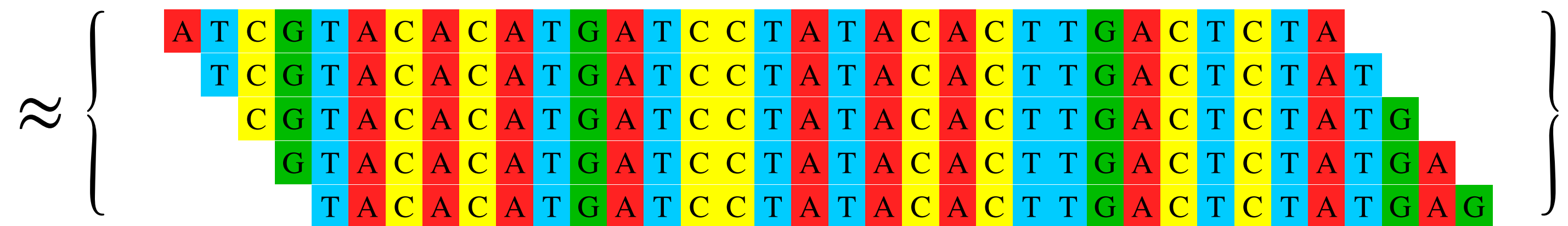
- The substrings of length k of a longer sequence:

A T C G T A C A C A T G A T C C T A T A C A C T T G A C T C T A T G A G

- 4-mers:



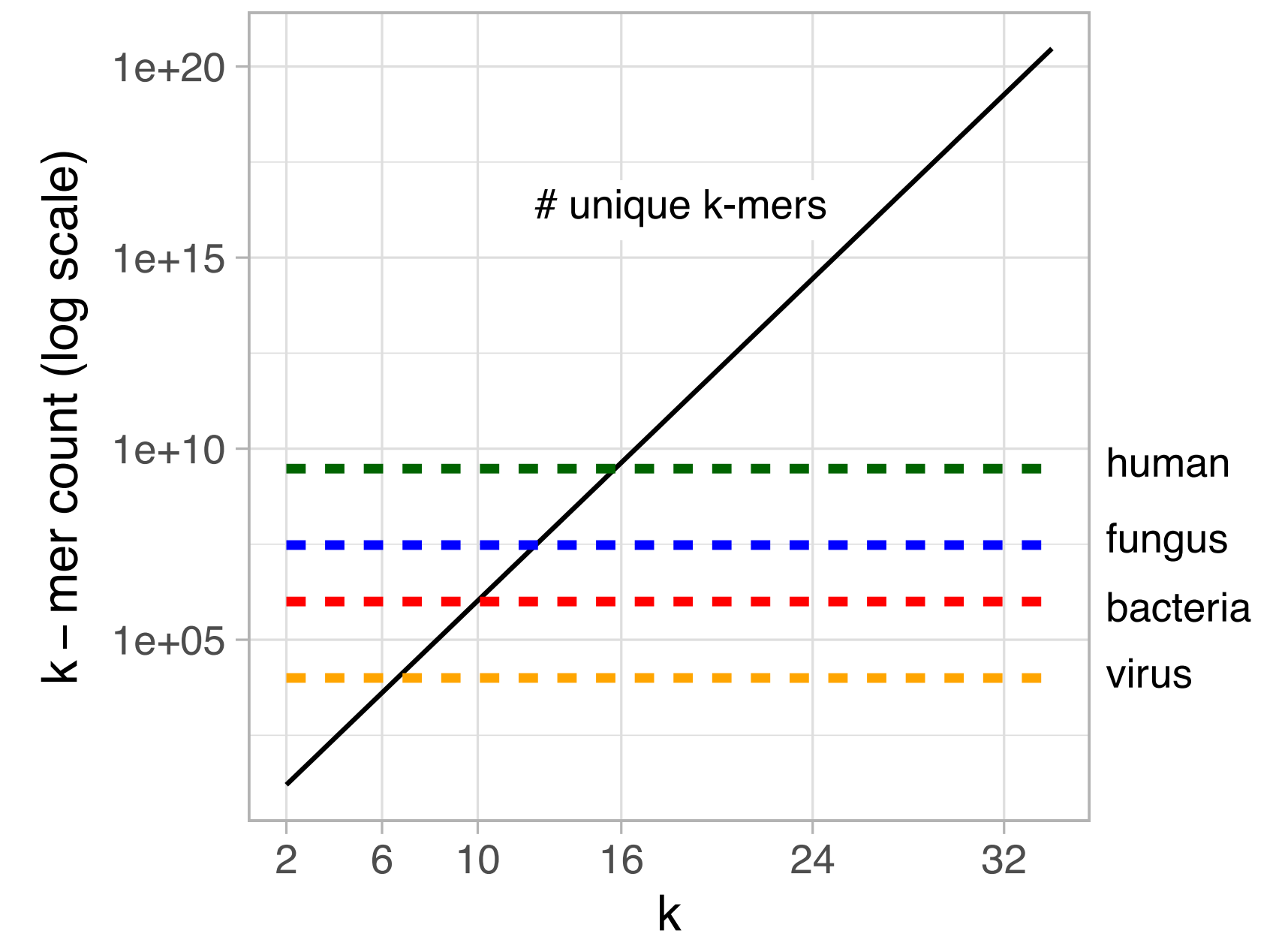
- 32-mers:



- — Represent a sequence as a **set** or **multiset** (not an ordered list)
- — Each k -mer can be represented as $2k$ -bit number $\Rightarrow$ a “long” for $k = 32$

How many k -mers are there?

- $L - k + 1$ k -mers in a string of length L
- 4^k possible k -mers (DNA)
- Are possible k -mers $>$ k -mers of a genome?
(do we expect k -mers in a genome to be unique?)

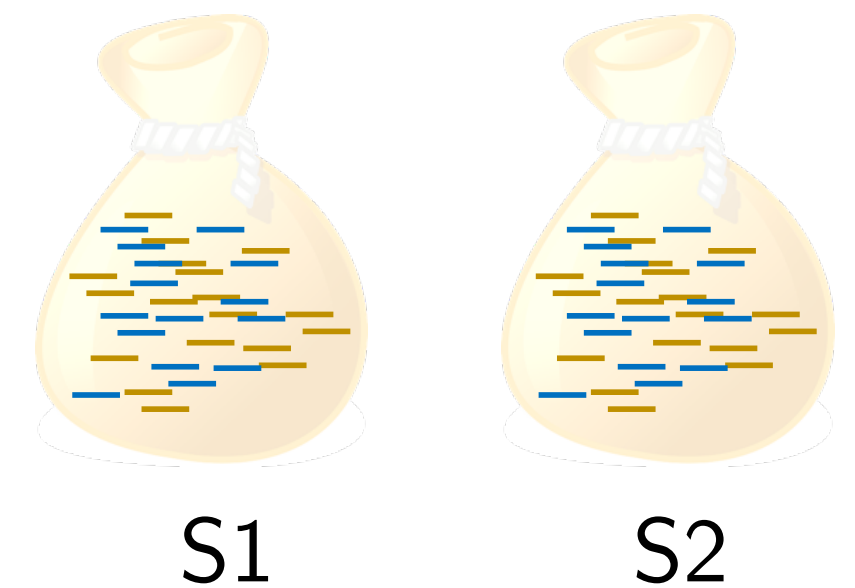


- Two randomly generated genomes of length $L = 4\text{Gbp} \approx 2^{32}$ share almost no k -mers. What k would lead to such a property?

- Remember the birthday paradox? $\sqrt{4^k} \approx L \Rightarrow k \approx 32$

k -mer representations: two paradigms

- View 1: A full sentence, found with low probability twice by chance alone!
- Bag of k -mer: **Presence/absence** of long k -mers ($k > 15$; often 20 – 32)
- View 2: A word, found in most/all (longish) texts
- Vectors: **Frequency** of all possible 4^k short k -mers (e.g., $k < 10$, often 3 – 8)



4^k

	AAA	AAC	AAG	...	TTT
S1	0.01	0.02	0.01	...	0.04
S2	0.05	0.01	0.03	...	0.02

Why take sequences to the butcher?

Power of k -mers for sequence comparison

1. **Scalability**: Easy-to-implement set operations on fixed-sized strings.
 - Powerful **hashing and sketching techniques** for efficient calculations from ultra-large data
2. **Simple math** under simplified yet useful models!
 - Often leads to analytical equations for parameters of interest
3. The choice of k gives us flexibility

Pairwise single-organism comparison

- Distance d = The fraction of sites that differ if two genomes are aligned (i.e., substitution probability)

G_1 ...ACCTGAAGGGTATCGCC...
 G_2 ...ACCTGAGGGGTATAGCC...

- Without alignment, represent each as a k -mer set

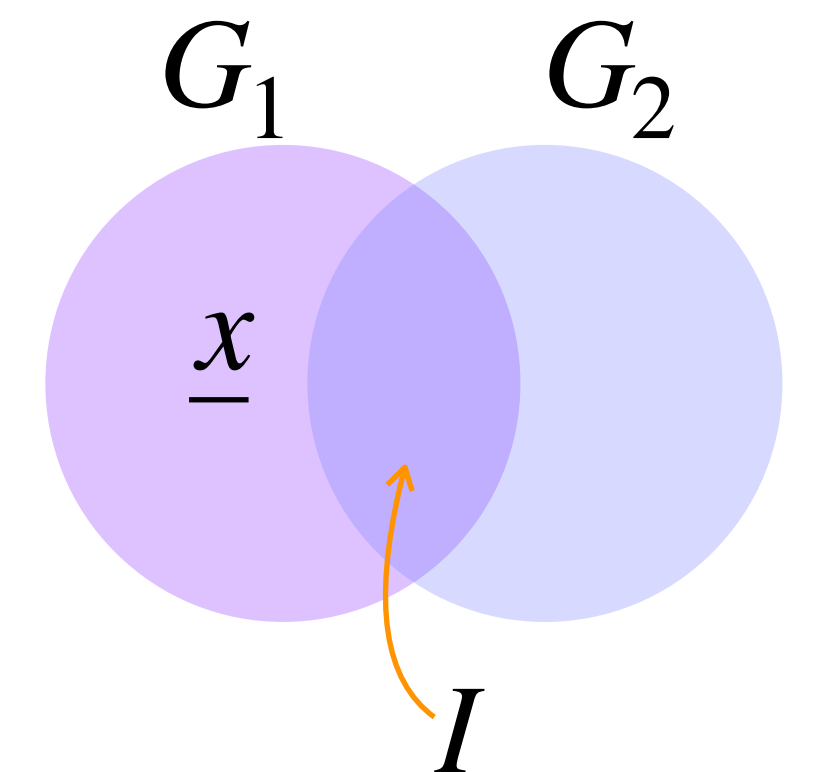
$$d = 1 - \text{ANI} = \frac{2}{16} = 12.5 \%$$

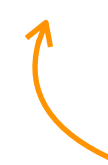
- For each k -mer x in G_1 :
 $P(x \text{ is found in } G_2) =$

Independence Ignore gains and losses!


$P(x \text{ is found in } G_2 \mid x \text{ has a homolog in } G_2) P(x \text{ has a homolog in } G_2) +$

$P(x \text{ matches a } k\text{-mer from a non-homologous position in } G_2)$



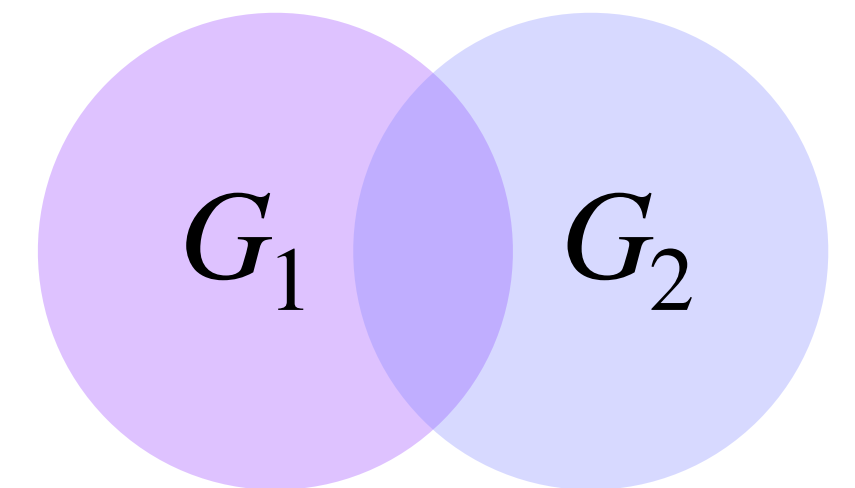

 Easy to measure $|I|$ empirically using set operations!

Can be made even more efficient with sketching

$$\bullet J = \frac{|I|}{|U|} \approx \frac{|I|}{2 \times |G| - |I|} \Rightarrow \frac{2J}{1+J} = \frac{|I|}{|G|} \Rightarrow$$

Unique k -mers
in each genome

$$\hat{d} = 1 - \left(\frac{2J}{1+J} \right)^{1/k}$$



$$J = \frac{|G_2 \cap G_1|}{|G_2 \cup G_1|}$$

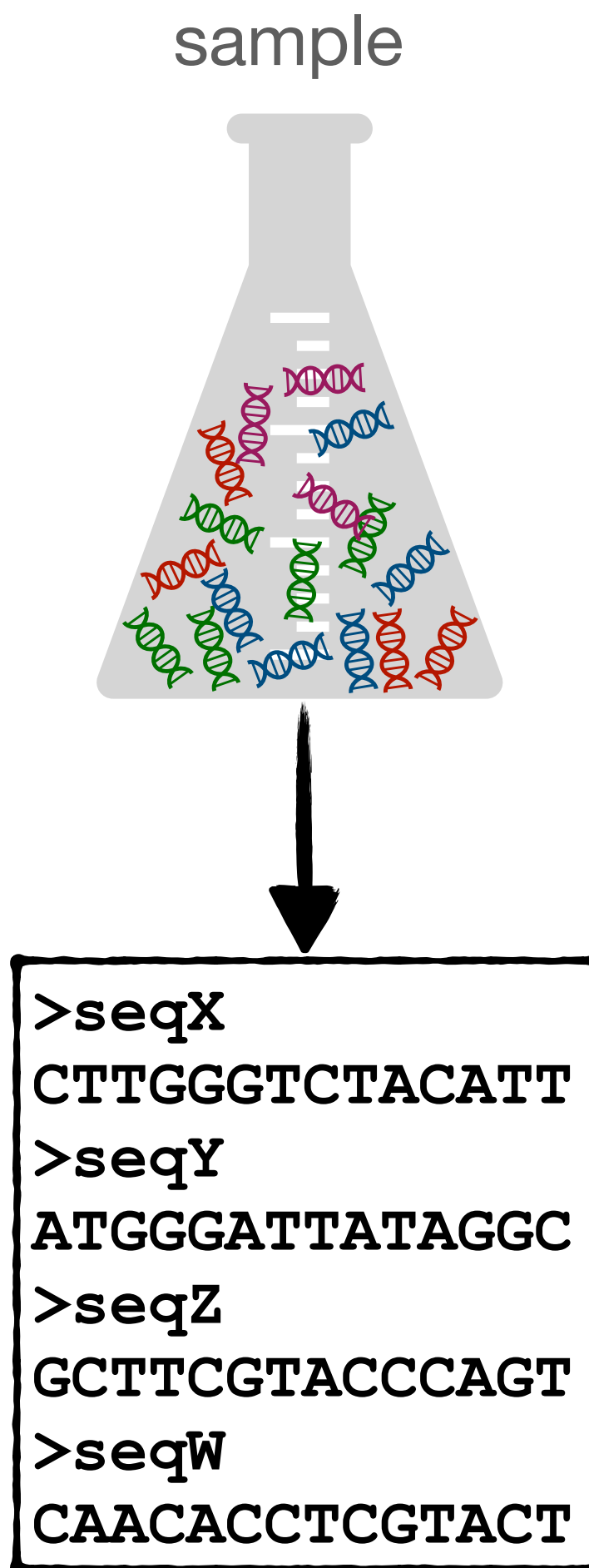
- Jaccard (J) can be calculated efficiently using a technique called **sketching** (e.g., MinHash in Mash and FracMinHash in Sourmash)
- Don't need all k -mers; **a small subset** suffices to compute J

What we focus on today

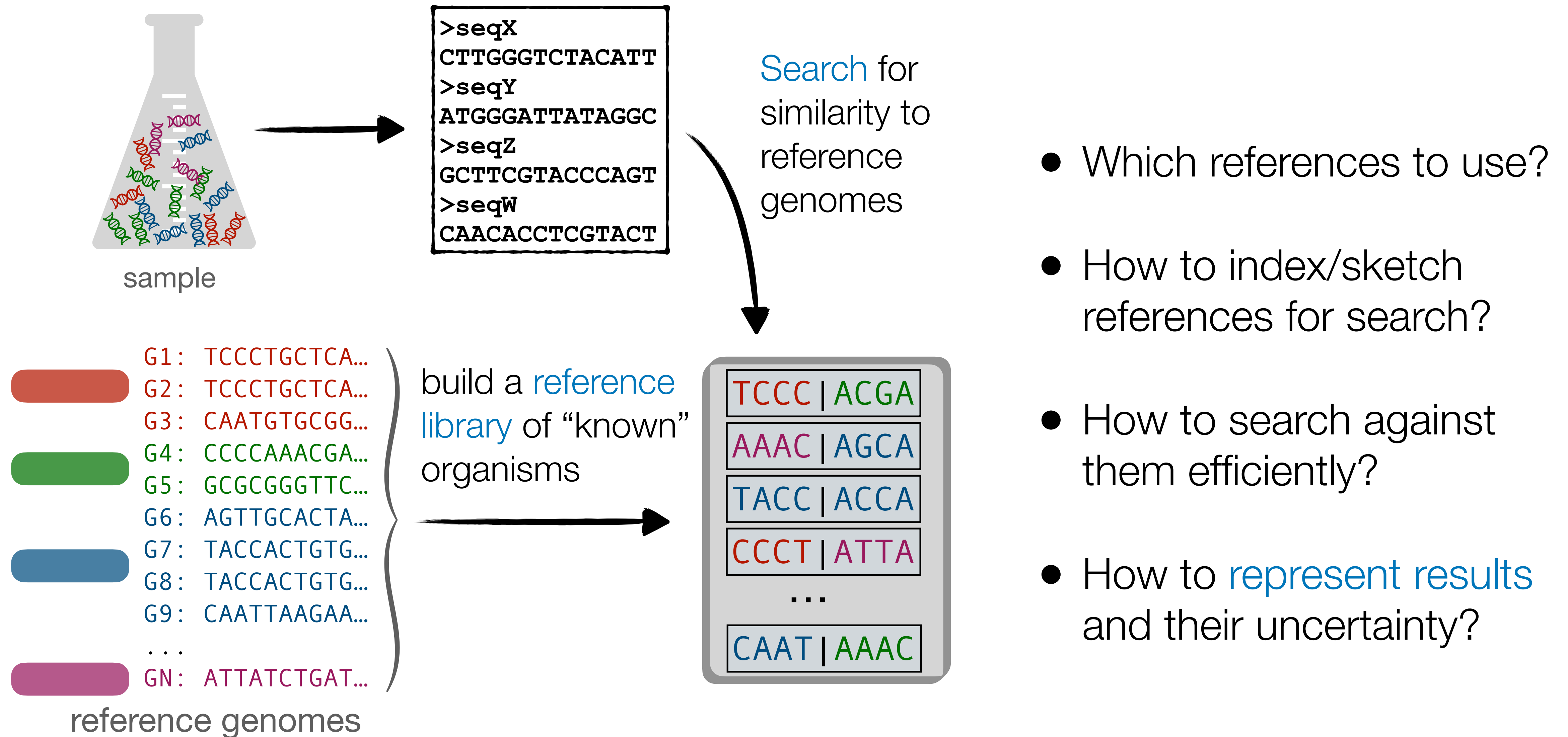
- k —mers used for genome assembly, estimating sequencing parameters (coverage, error), “seeding” alignments, tokenizing sequences, ...
- We focus on **sequence comparison**: Find similarities between sets of sequences (genomes, contigs, sets of reads, ...)
 - A. **Single-organism input**: Compare a pair of inputs, each presumably from a single organism (**not our focus!**)
 - Mash, MashMap, CAFE, Skmer (ReSkmer, DipSkmer), skani
 - B. **Mixture analyses**: One or both inputs are mixtures of organisms

Mixture sequence data

- Sequence a “sample” obtained from a mixture of organisms.
 - We always have DNA from multiple origins. Sometimes this is not a bug (contamination) but a feature.
 - **Cannot separate** constituents easily (perhaps not at all).
 - **The composition** is of interest.
- Two types (roughly):
 - **Metagenomics**: shotgun sequencing of all DNA (genome-wide data)
 - **Targeted** sequencing of a marker gene (e.g., 16S) or a portion of it (ASV)

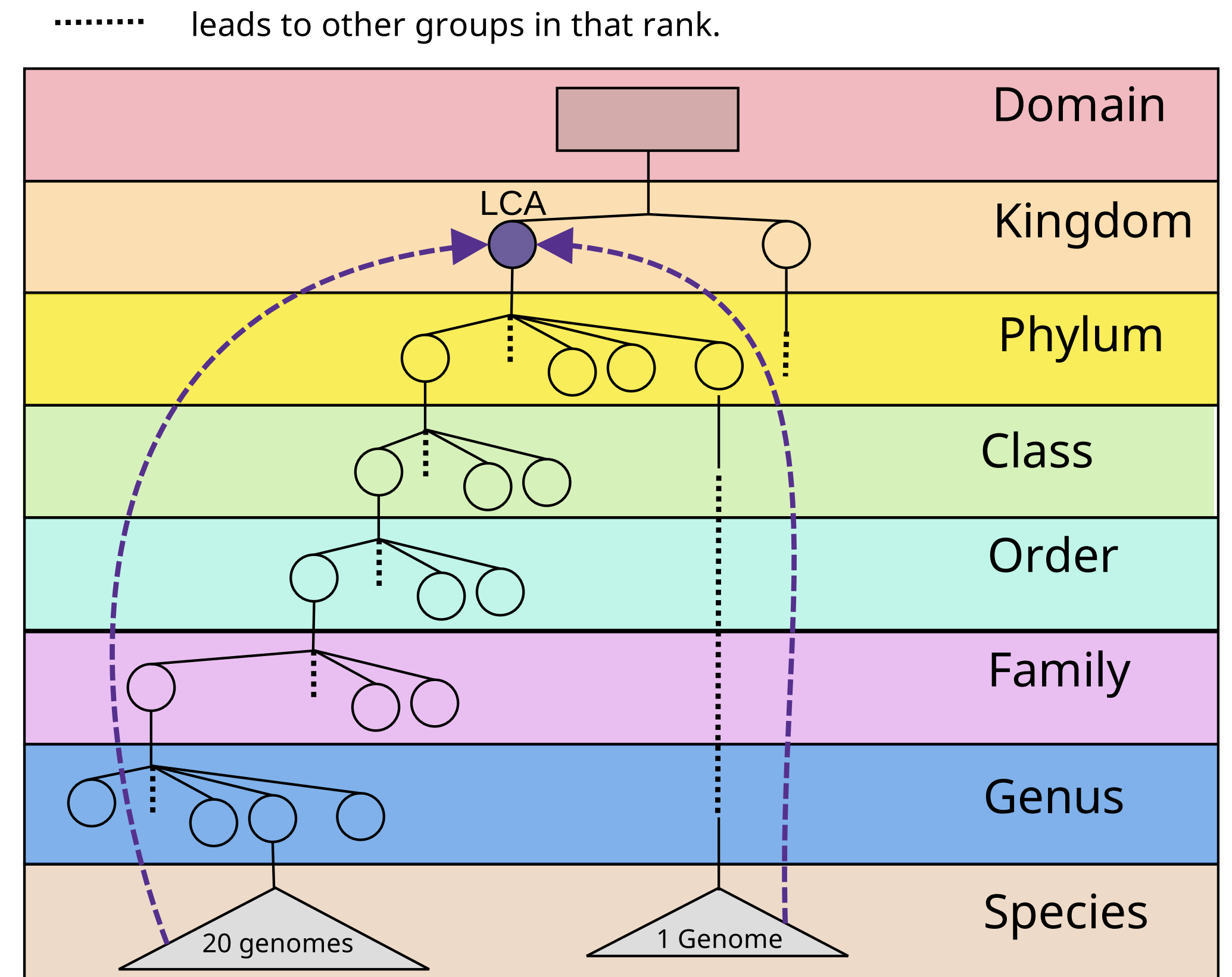


Finding constituents: comparison to references

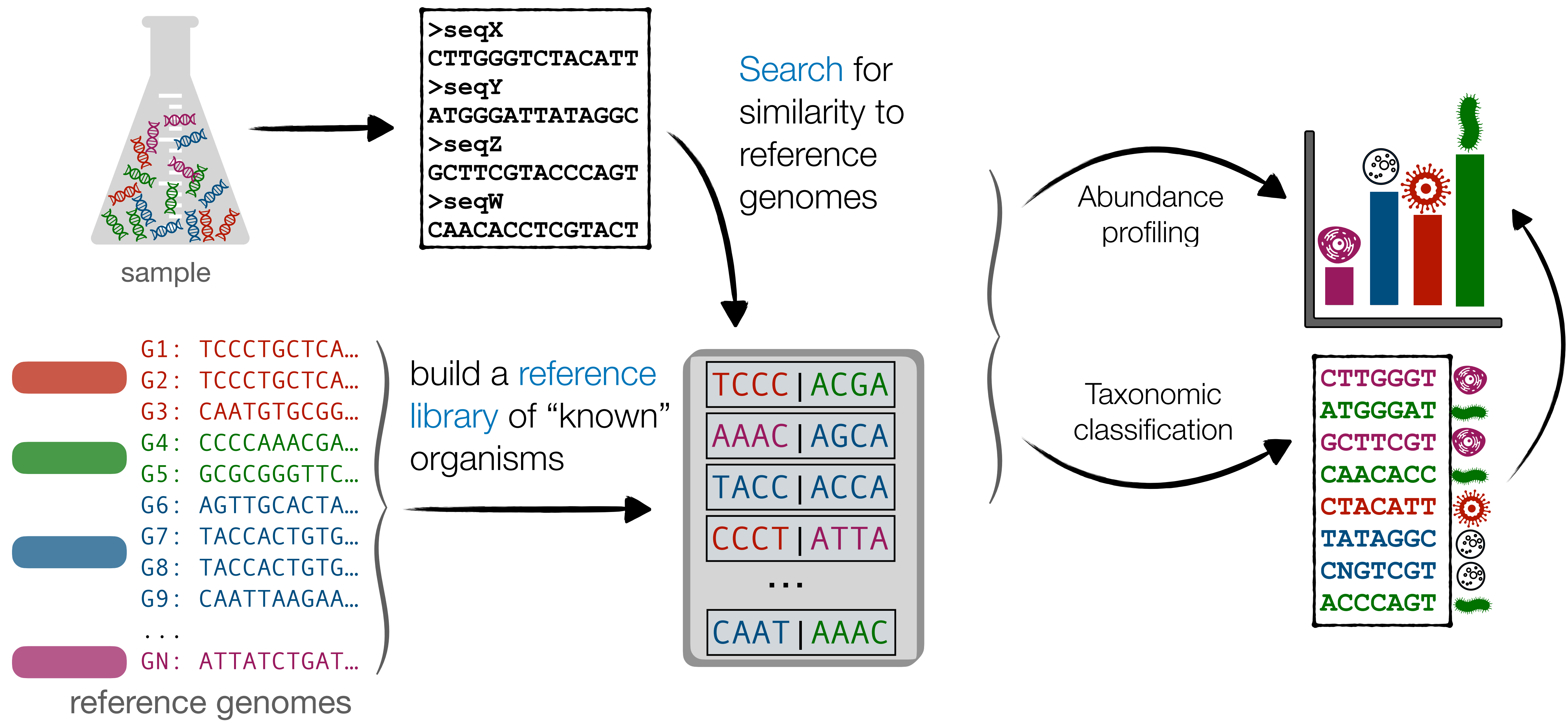


Taxonomical hierarchy

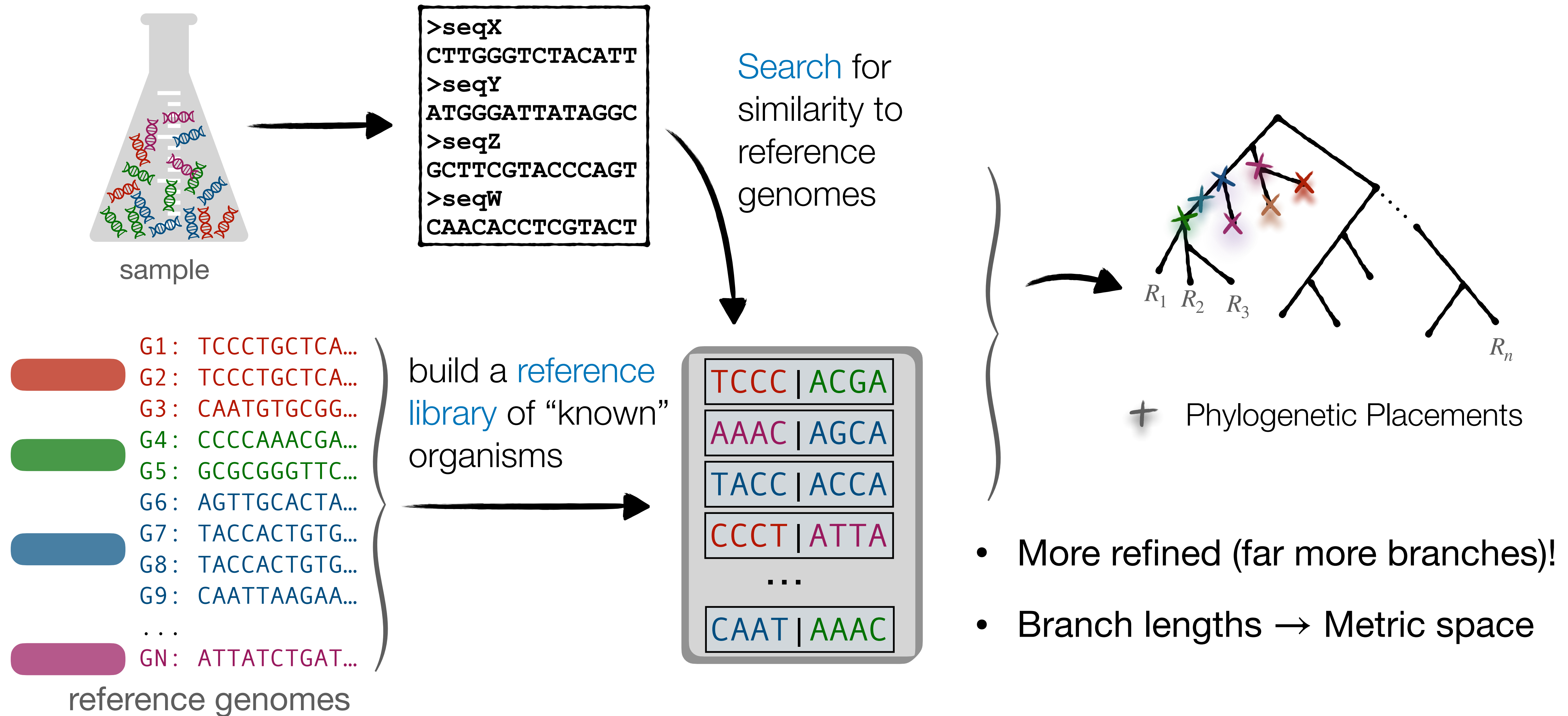
- A **naming system**, organized as a tree
 - Available for reference datasets
 - Enables a **hierarchical classification** problem, going to higher levels when uncertain
- Broadly (but not exactly) following phylogenetic relationships; however,
 - **Highly multifurcating** (e.g., 2.6K vs 15.9k = 16% of a phylogeny)
 - **No branch length**



Finding constituents: comparison to references



Finding constituents: placing on a reference phylogeny



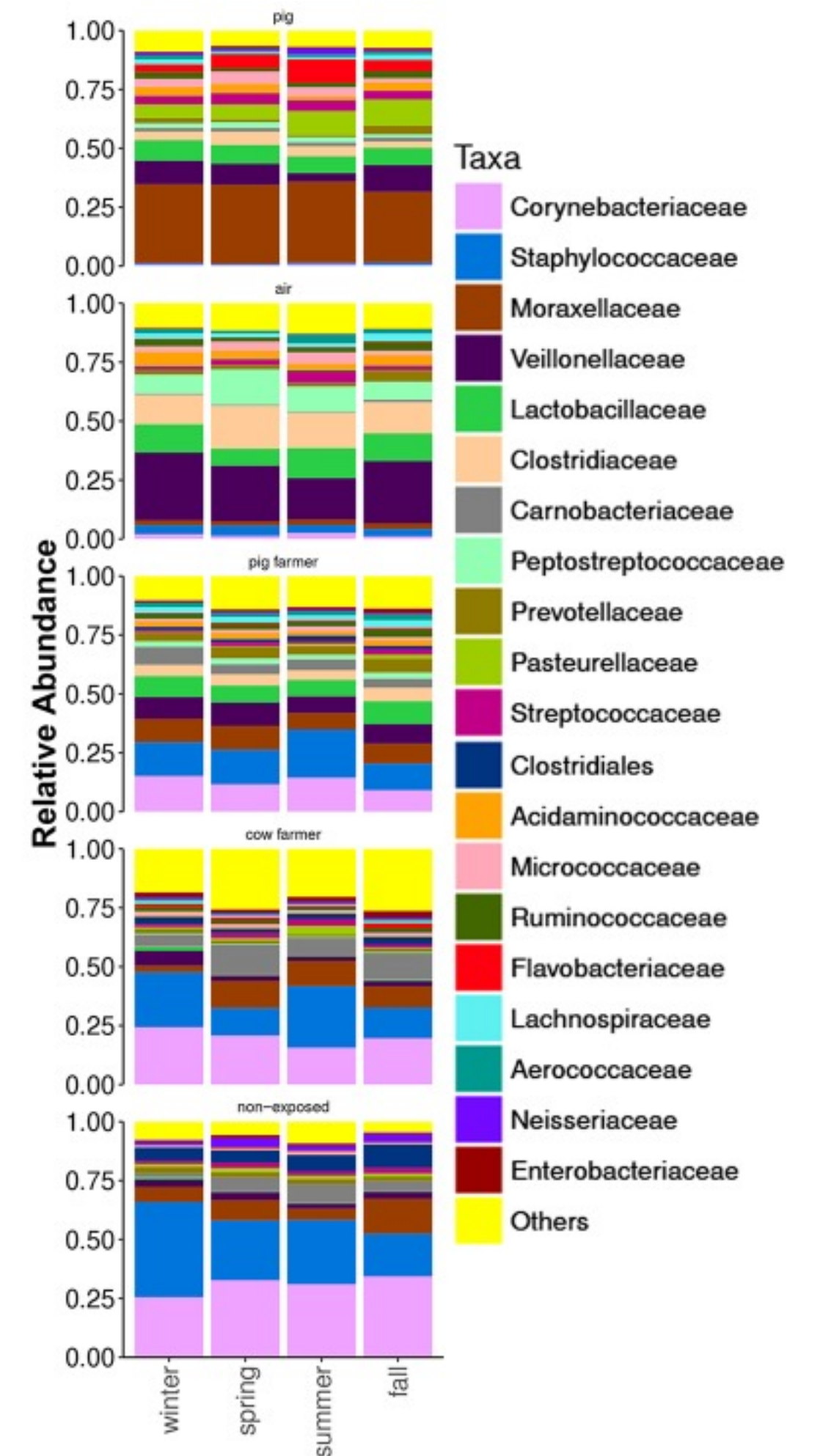
**What do we do with the
results?**

Detection: Is the organism x present in the sample?

- Should we detect even at low abundance?
- How about similar organisms — are they “the same”?
- Needs statistical testing

Get many samples and **compare** them

- across time points!
- across host conditions (e.g., healthy/disease)
- across environments (e.g., deep sea, shallow sea)
- in response to stimuli (e.g., treatments)
- in terms of (alpha) diversity



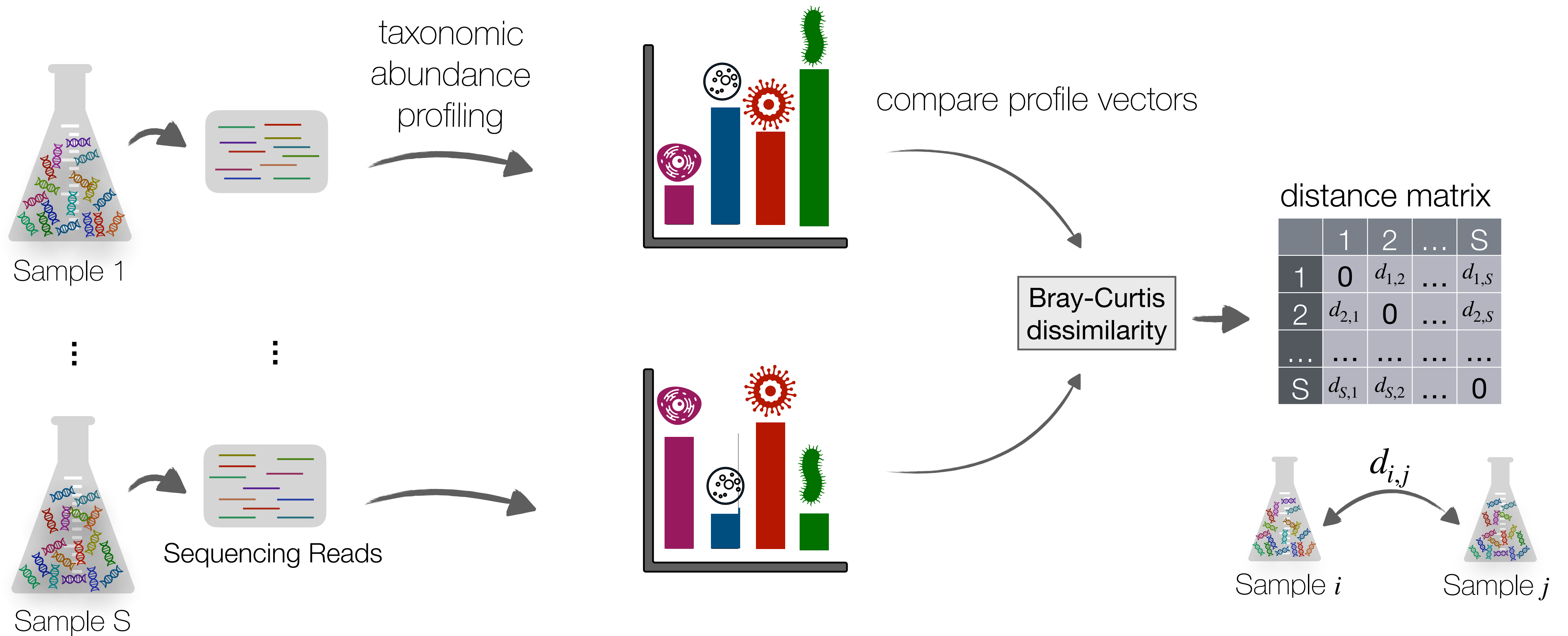
Applied and
ASM Environmental Microbiology

The Indoor-Air Microbiota of Pig Farms Drives the Composition of the Pig Farmers' Nasal Microbiota in a Season-Dependent and Farm-Specific Manner (2019)

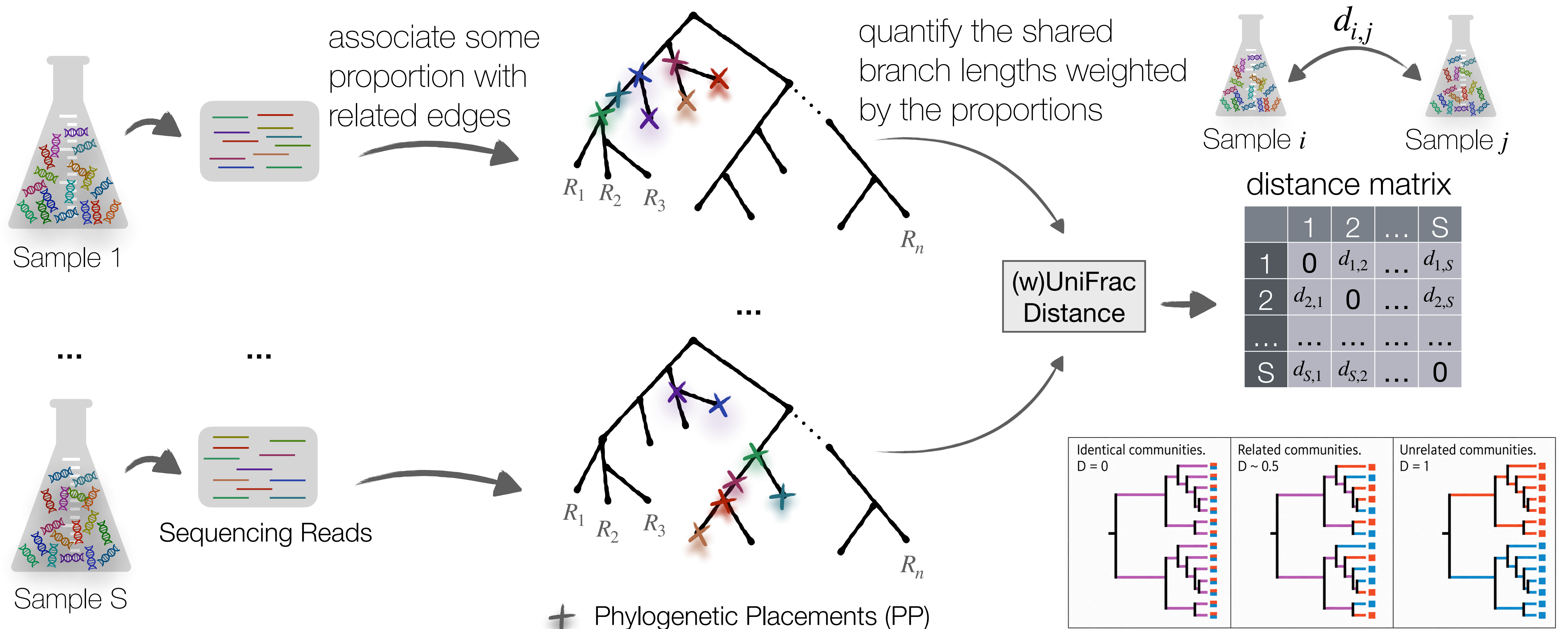
[Julia G Kraemer](#), [Susanne Aebi](#), [Anne Oppliger](#), [Markus Hilty](#)

... bacteria from the noses of pigs are more commonly coidentified within the pig farmers' microbiota in winter but to a less extent in summer ...

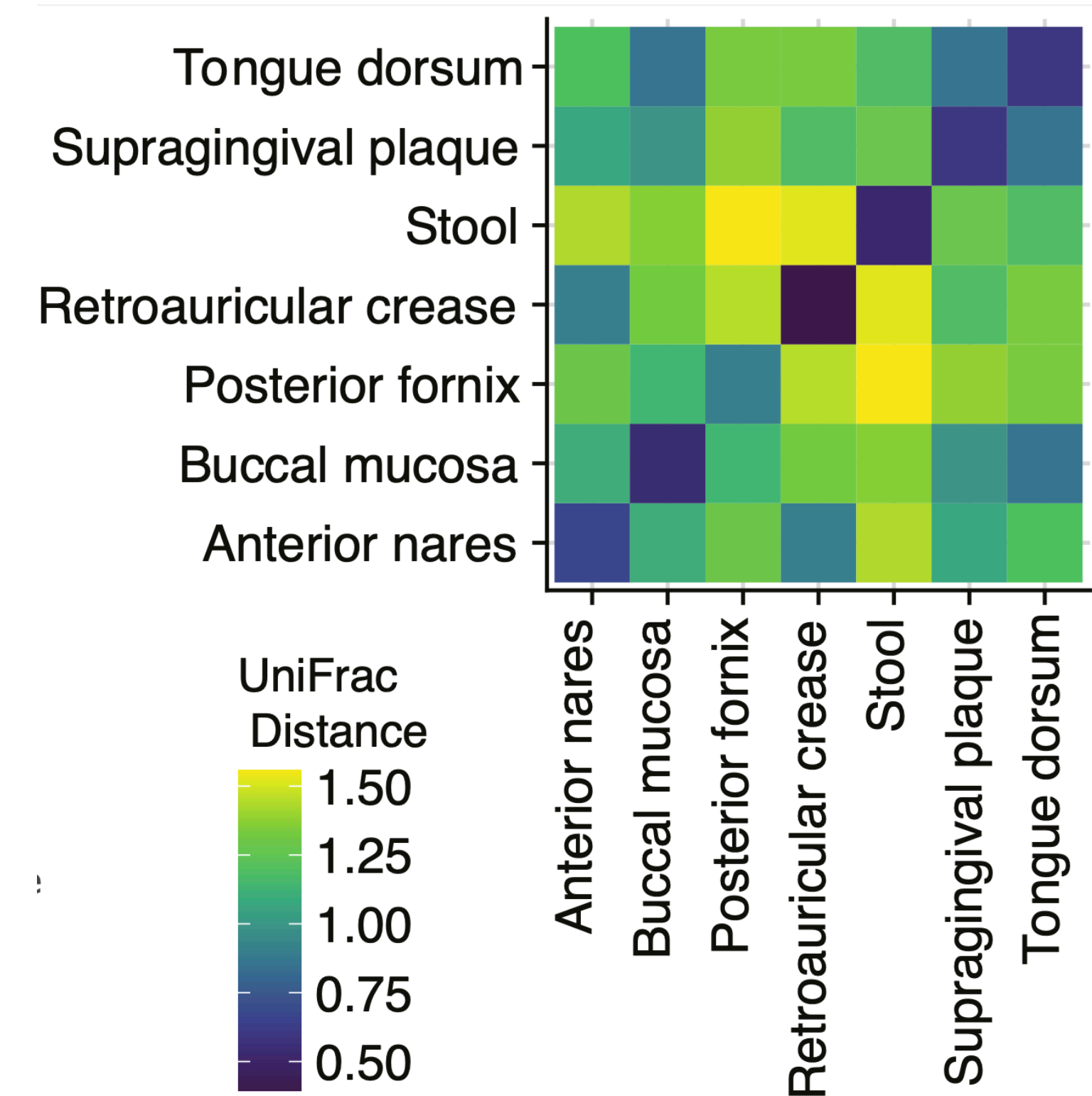
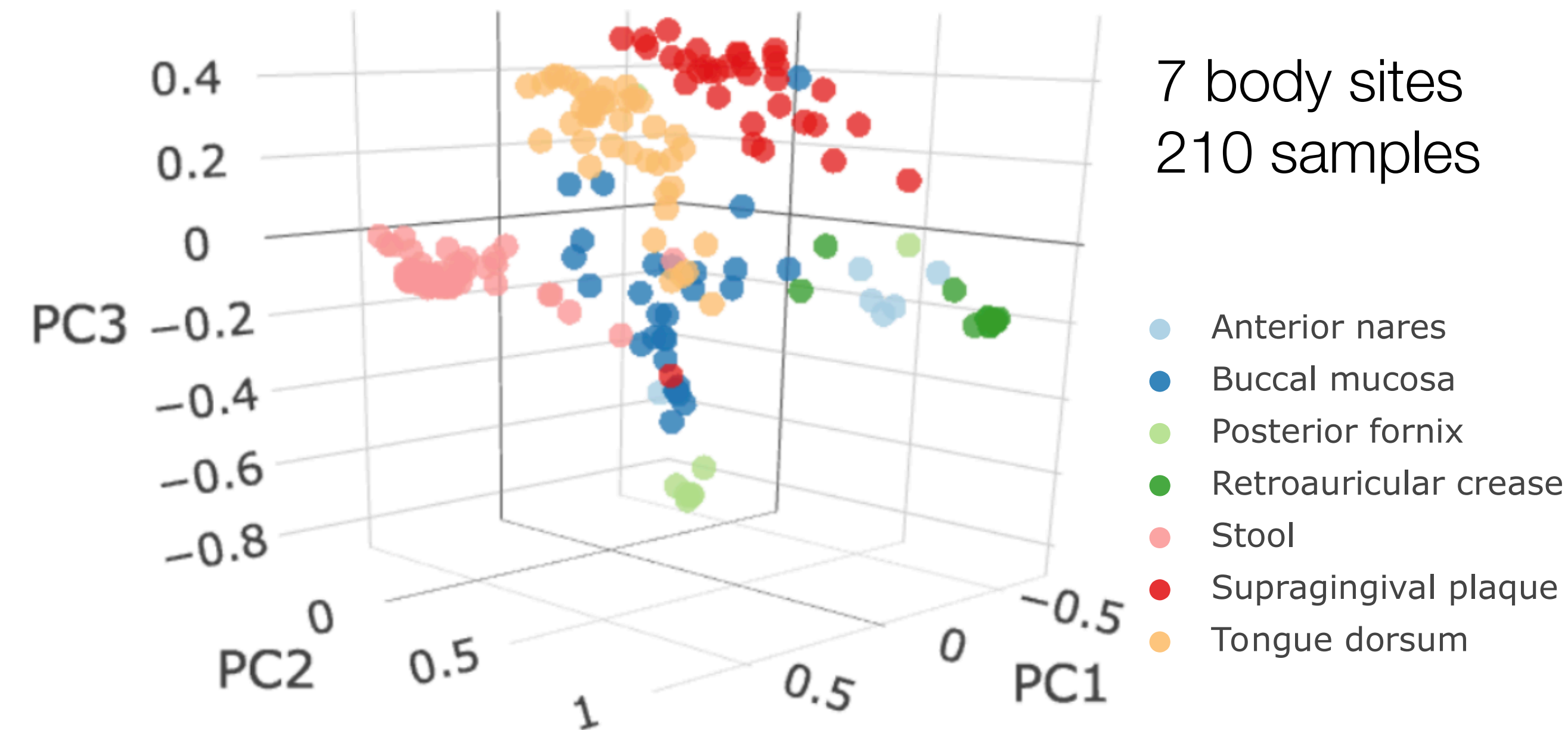
Comparing samples after profiling



What if we also have a reference phylogeny?



Better resolution with alignment and UniFrac

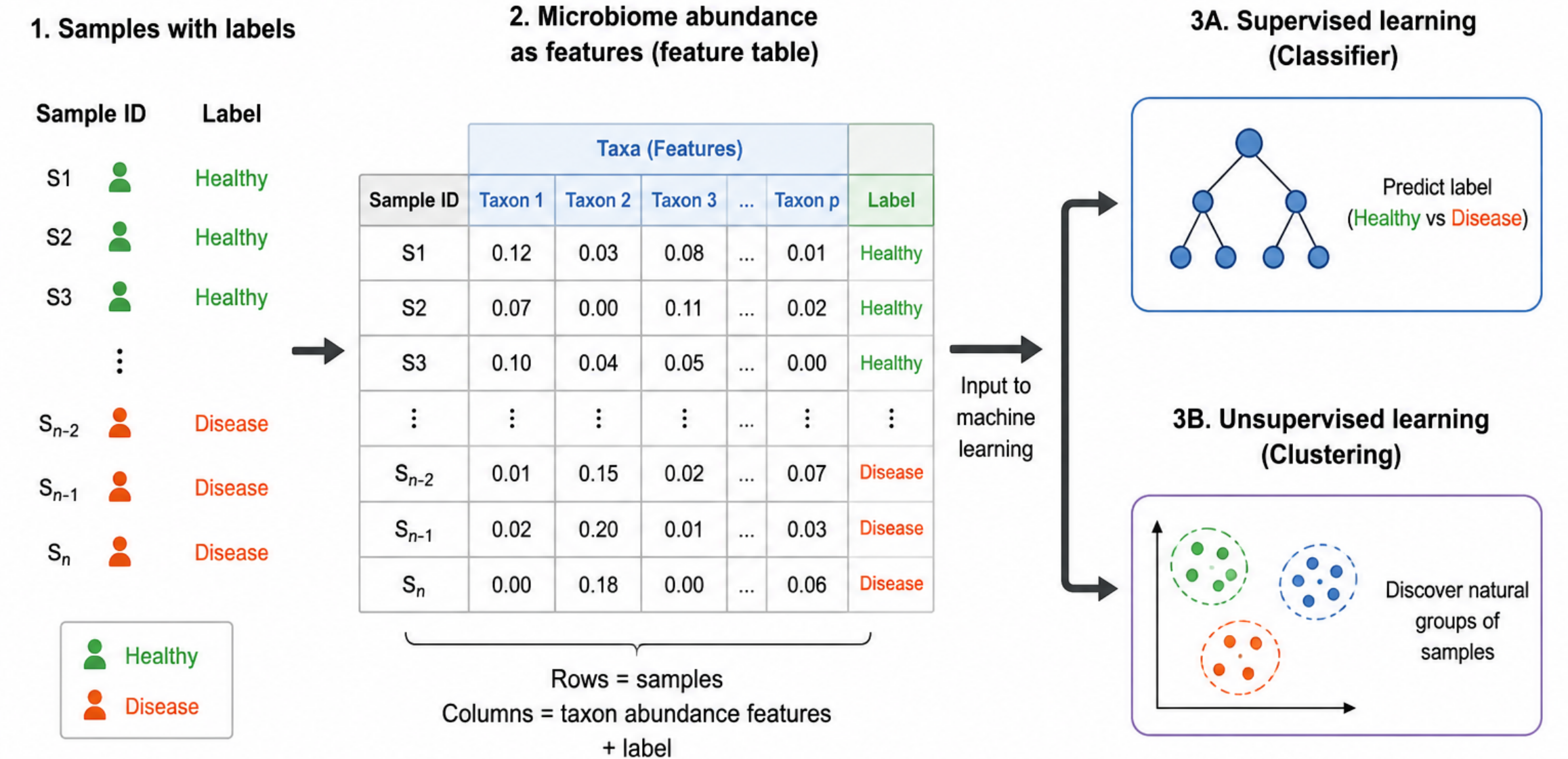


Visualize using MDS ($\approx$ principal coordinate analysis; PcoA): Find d -dimensional points with distances matching pairwise sample distances

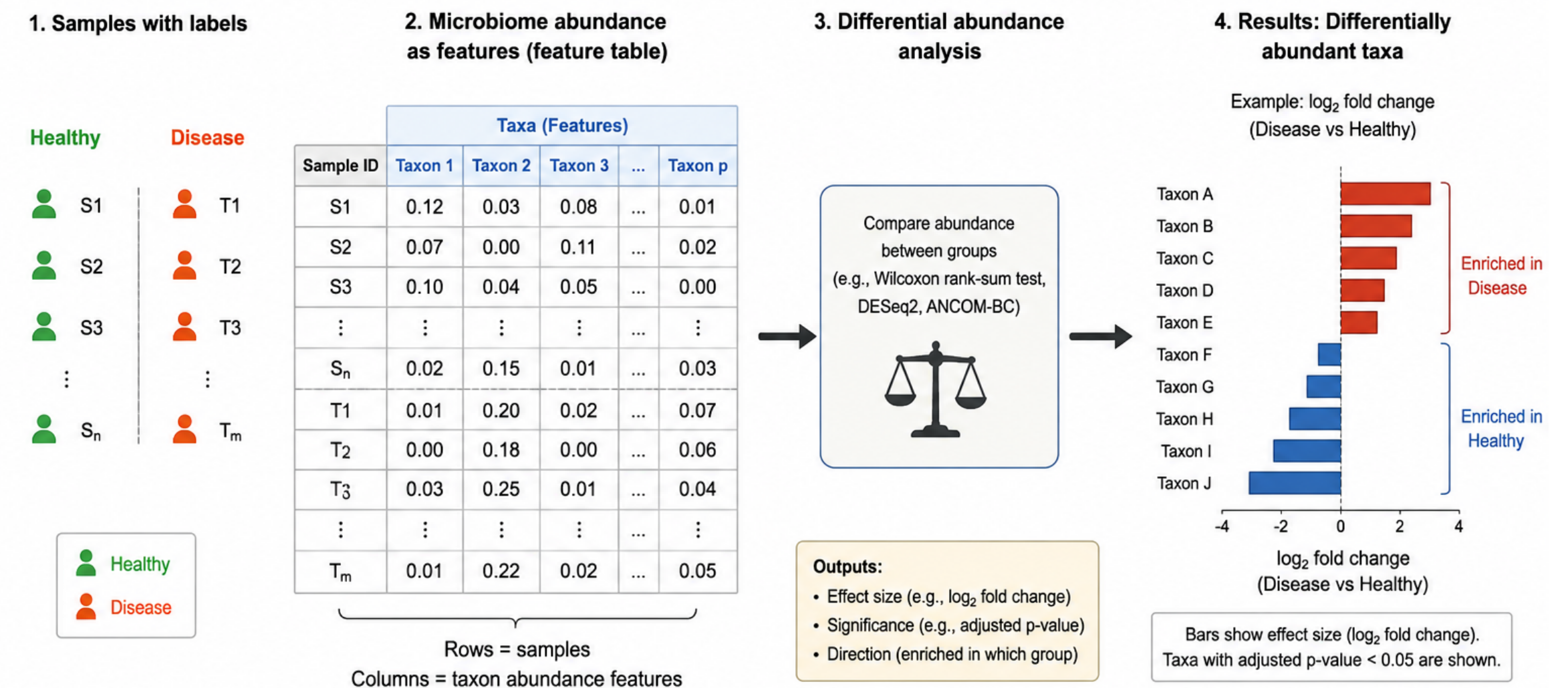
Measure associations of distances to classes:
pseudo- F statistic: Across/within group distances

Use taxa as “features”

- Train **machine learning models** to predict labels from taxonomic profiles or cluster them



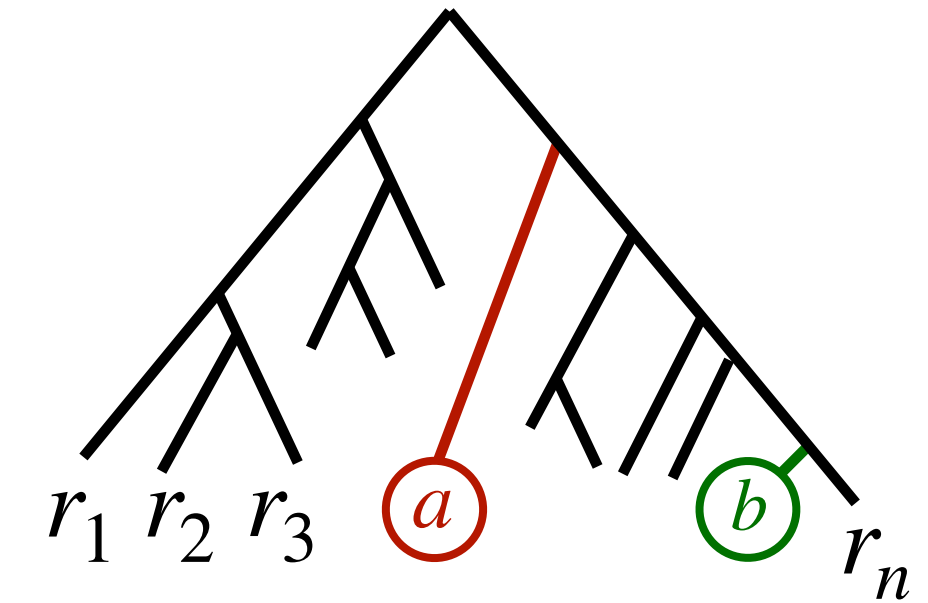
- Use statistical analyses to quantify changes in abundance (e.g., **differential abundance**)



Challenges: computational scalability

- Large reference datasets (hundreds of thousands of genomes)
- Large numbers of samples
- Large numbers of reads per sample

Challenge of “novel” constituents



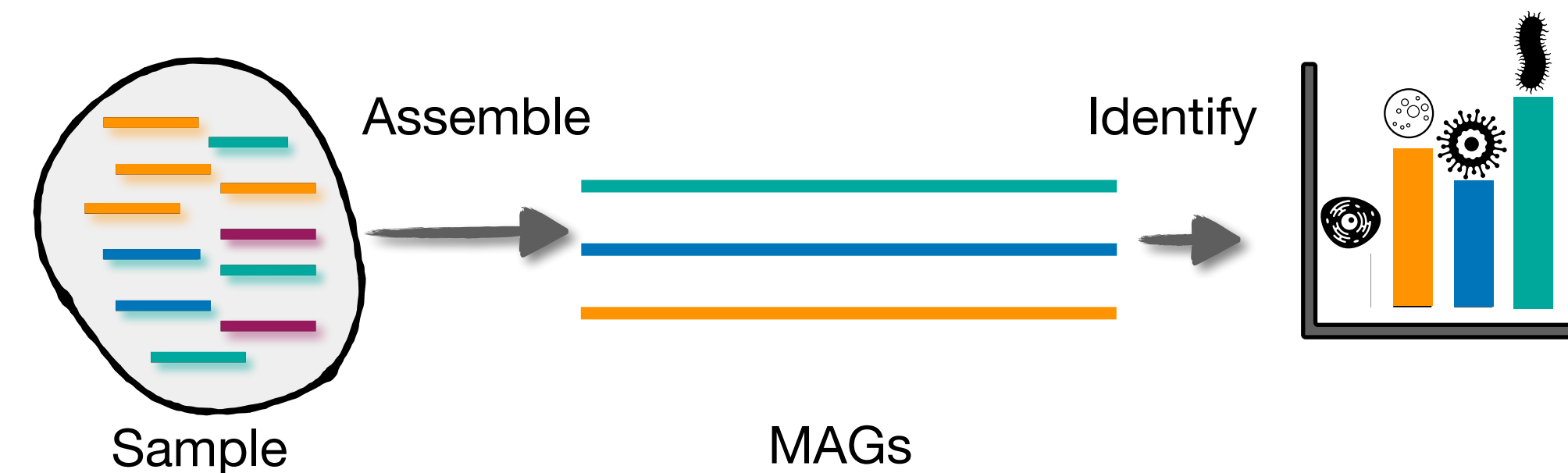
- “**Novel**”: not similar to any of the references we happen to have!
- Novelty has levels; e.g., the ANI to the closest matches may be **low** or **high**
 - Only 12% of Earth’s microbiome has >97% ANI matches in a reference set [Zhang et al. 2020]. Remaining 88%: novel sequences!
 - Novelty at <90% ANI is also routine for many microbiome data

Complex compositions

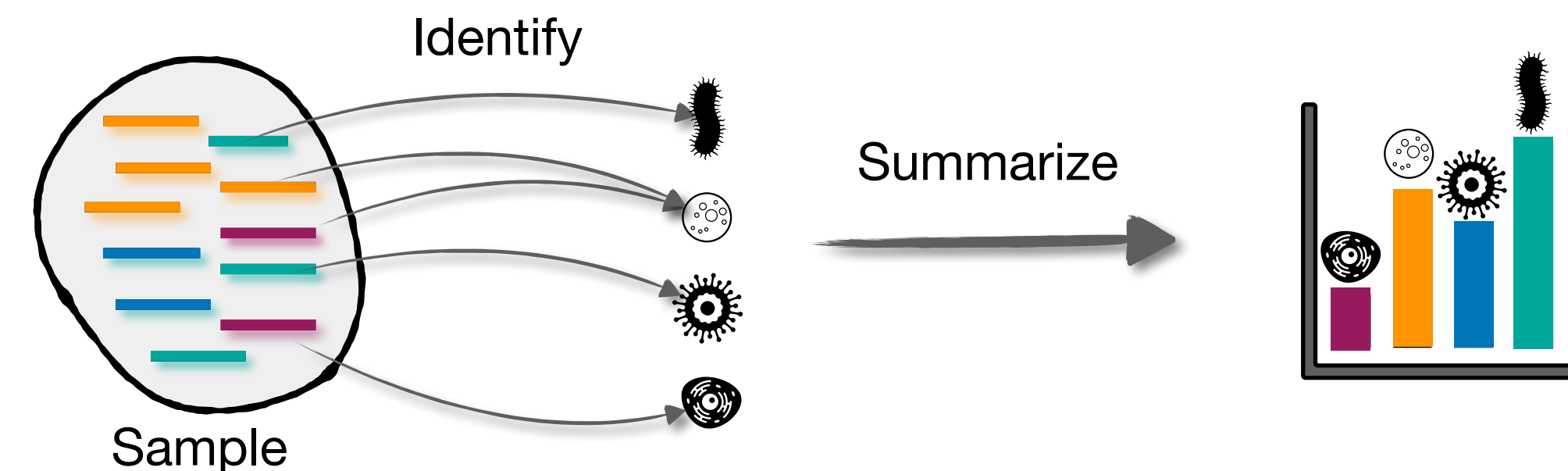
- Unknown **number of constituents**
- Abundances can range from uniform (easy!) to **highly variable (difficult!)**
- Some constituents may be very **similar to others** (strains of a species)
- Cell count versus sequence abundance

Metagenomic data: common pipelines

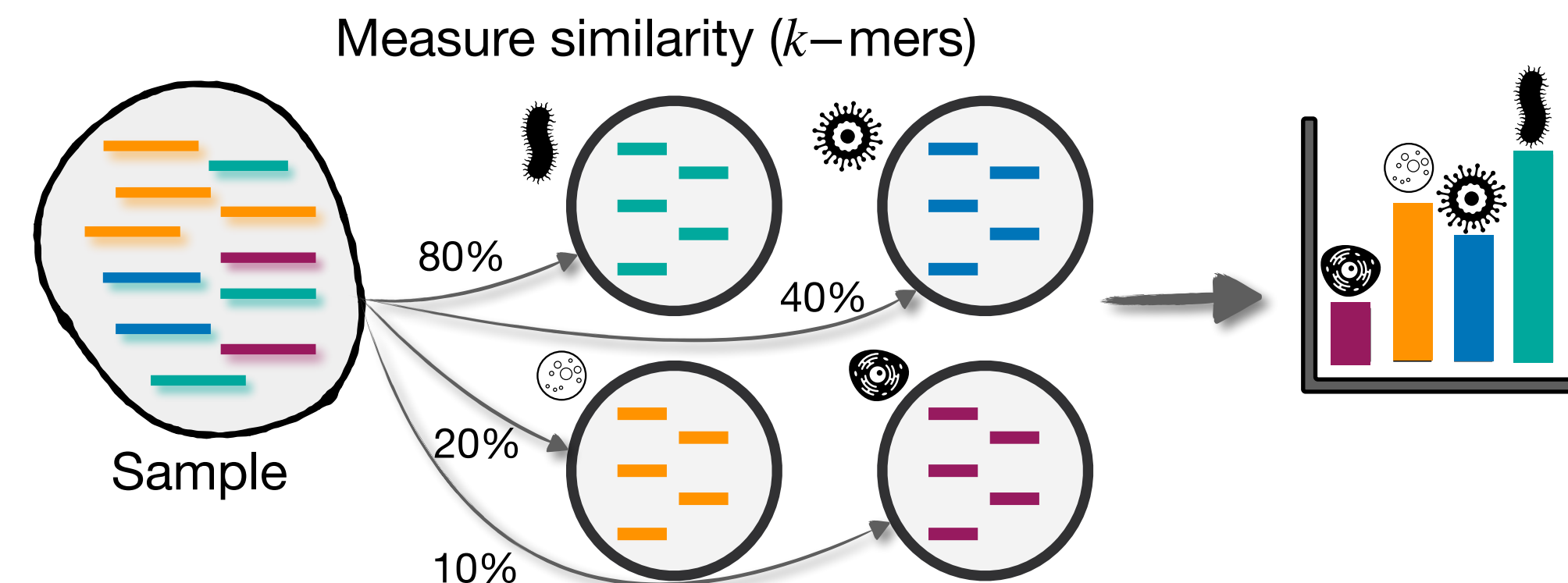
- Metagenome Assembled Genomes (MAGs):
 - 1) Assemble reads into putative genomes.
 - 2) Identify those genomes



- Two-step identification:
 - 1) Analyze **each read** to identify it
 - 2) **Summarize** results across reads



- Whole sample (**joint**) identification:
 - 1) Compare the sample as a whole against the references
 - 2) Aggregate results



Which is better?

- MAG:
 - Longer sequences → better signal
 - Unassembled parts left out
 - Slower, more involved pipelines
- Two-step (per read):
 - All reads considered → Sensitive
 - Convenient and scalable pipelines
 - Noisy estimates per read → noisy summary
- Joint analysis:
 - All reads considered
 - No per read noise
 - Convenient+Scalable
 - Tend to struggle with novel sequences (?)

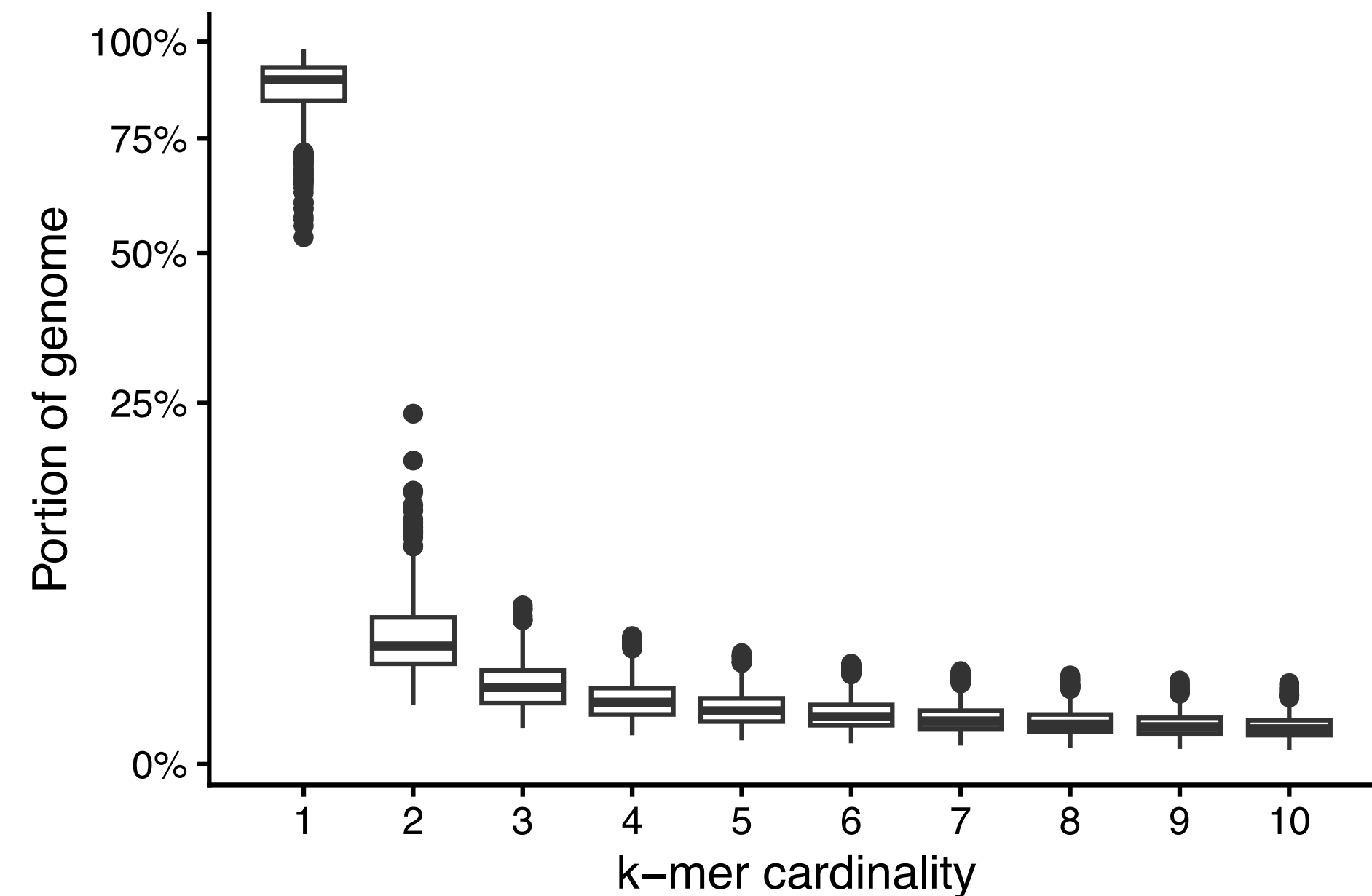
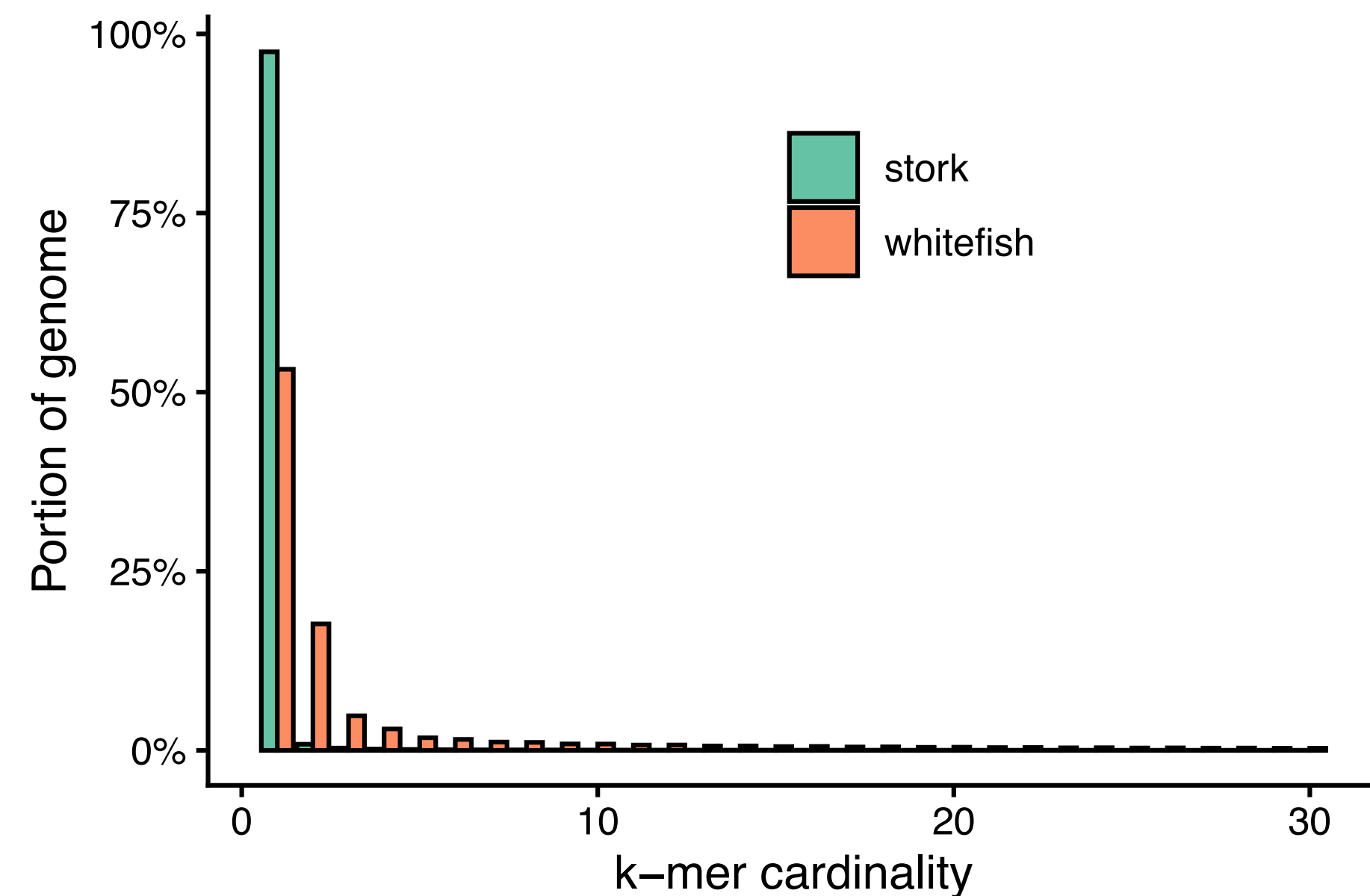
We will cover much of it!

	Kraken-II	Krepp	Sourmash	Yacht	Sylph	Skani	kf2vec	virFinder	Decodiphy
Long/ Short <i>k</i>	Long (<i>k</i> = 31)	Long (<i>k</i> = 29)	Long (<i>k</i> = 21–51)	Long (<i>k</i> = 31)	Long (<i>k</i> = 31)	Medium (<i>k</i> = 15)	Short (<i>k</i> = 7)	Short (<i>k</i> = 7)	-
Input	Per-read + summary	Per-read + summary	Whole sample	Whole sample	Whole sample	MAG / Genome	MAG / contigs/ Long read	Contigs	Whole sample
<i>k</i> -mer search	Minimizers + Masked hash	Minimizers + LSH	FracMinHash sketching	FracMinHash sketching	FracMinHash sketching	FracMinHash + positional hash maps	<i>k</i> -mer counting	<i>k</i> -mer counting	-
Output granularity	Taxonomy	Phylogeny	Taxonomy + Containment	Taxonomy+ Species presence test	Taxonomy + ANI	ANI + alignment coverage	Phylogeny	Viral vs. host	Phylogeny
Algorithm	LCA over <i>k</i> -mer matches	ML distances + placement	Containment Jaccard + Set cover	Containment Jaccard + hypothesis	Containment- based ANI + coverage	Containment ANI + sparse chaining	Machine learning	Machine learning	Phylo. Distance Deconv.

Questions?

Are 32-mers actually unique across the genome?

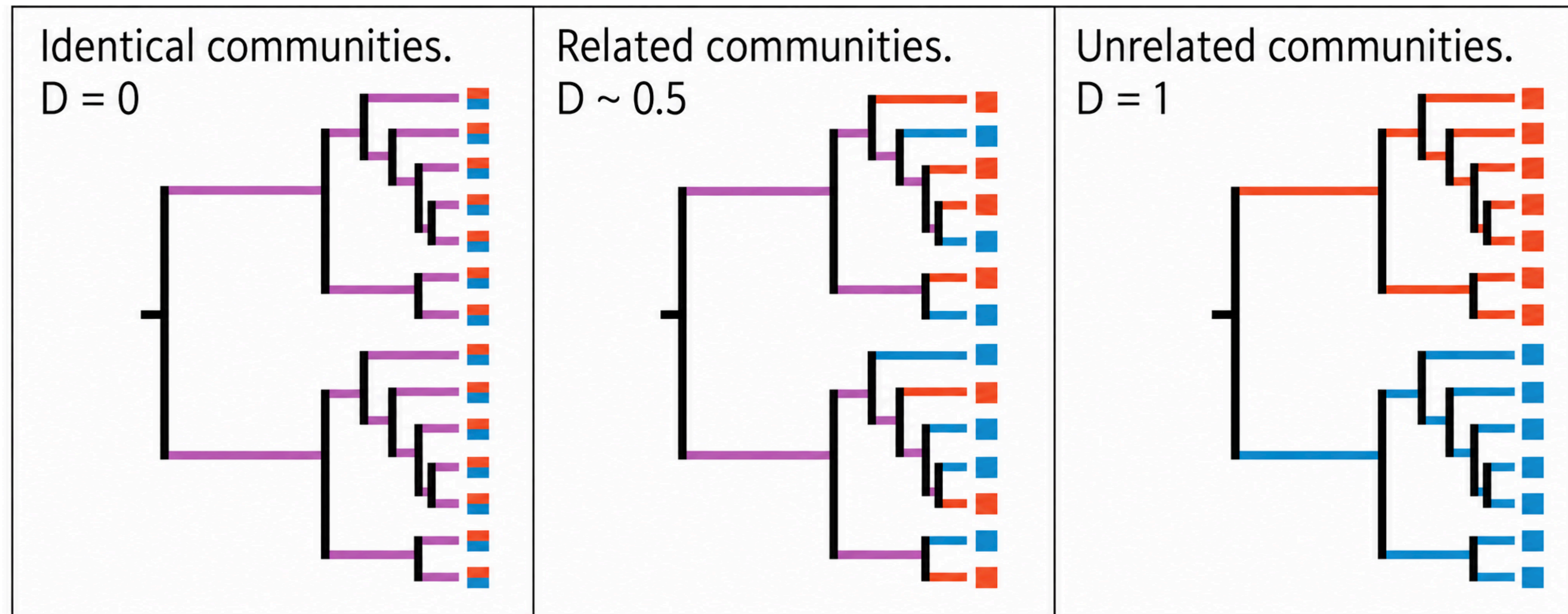
k -mer frequency spectrum



Across 570 vertebrates

- Depends on the species!
- Portion of unique 32-mers can be as low as 30 — 40% in some species

Unique Fraction (UniFrac) metric



[Lozupone et al. 2005]

Let b_i be the length of the branch i and p_i^A and p_i^B are the taxa proportions descending from the branch i for community A and B , respectively.

wUnifrac:

$$d(A, B) = \frac{\sum_i^n b_i |p_i^A - p_i^B|}{\sum_i^n b_i (p_i^A + p_i^B)}$$

Pseudo-F statistics

$$SS_T = \frac{1}{N} \sum_{i=1}^{N-1} \sum_{j=i+1}^N d_{ij}^2 \quad SS_W = \sum_{g=1}^p \frac{1}{n_g} \sum_{i=1}^{n_g} \sum_{j=i+1}^{n_g} d_{ij}^2 \quad SS_A = SS_T - SS_W$$

$$F = \frac{\left(\frac{SS_A}{p-1} \right)}{\left(\frac{SS_W}{N-p} \right)} \xrightarrow[\substack{\text{Fix } n = n_1 = n_2 \\ d_w = \text{within } d_a = \text{across} \\ \text{distances}}]{\text{ }} F = n \left(\frac{d_a^2}{d_w^2} - 1 \right) + 1$$

Multiple permutations to get a p -value!

